Callex — Production QA & Server Restart Guide

> ***Version:*** *1.0* | ***Date:*** *March 17, 2026* | ***Prepared by:*** *Callex Engineering*

The Problem — Why APIs Return "Invalid or expired token"

The error `{"error": "Invalid or expired token"}` means the **server has not been restarted** after the latest code was pulled.

The new code fixes are on GitHub but the server is still running the **old code** in memory. You must restart the server every time code is pulled.

Step 1 — SSH Into the Server

```
ssh root@62.171.170.48
```

Navigate to the project directory:

```
cd /usr/src/sumit/elevenlabs/freeswitch-elevenlabs-bridge
```

Step 2 — Pull Latest Code (Do This Every Time)

```
# Discard any local server changes and pull fresh
git stash
git pull origin main
```

Expected output:

```
Already up to date.
```

or

```
Updating abc1234..def5678
 enterprise/backend/src/routes/analytics.js  | 99 ++++
 enterprise/backend/src/routes/external.js   | 23 ++
 app/main.py                                  | 76 +-
```

Step 3 — Restart All Services

```
pm2 restart all
```

To verify they restarted:

```
pm2 list
```

You should see all processes with status `online` and a fresh uptime (seconds, not hours).

If you only want to restart the Node.js backend:

```
pm2 restart enterprise-backend
```

If you only want to restart the Python voice engine:

```
pm2 restart callex-python
```

> ***Check `pm2 list` to see the exact process names on your server.***

Step 4 — Verify the Server is Running

```
curl http://localhost:4500/api/health
```

Expected response:

```
{"status": "ok", "ts": "2026-03-17T13:30:00.000Z"}
```

Step 5 — Test Both APIs in Postman

API Key to use (yours):

```
ck_live_d3188b50_94a44765acfc97d19ced4416
```

TEST 1: Identity Check (Run This First)

`GET http://62.171.170.48:4500/api/v1/debug/my-identity`

Header:

```
Authorization: Bearer ck_live_d3188b50_94a44765acfc97d19ced4416
```

What to check in the response:

```
{
  "userId": "abc123",
  "ownedAgents": 3,
  "agentIds": ["agent1", "agent2", "agent3"],
  "callsWithUserId": 5
}
```

Field	What it Means
`ownedAgents`	Number of AI agents linked to this API key. Must be > 0
`callsWithUserId`	Calls directly tagged with this userId

> ⚠️ ***If `ownedAgents` is 0*** *— your API key belongs to a different user account than the one that has agents. This is a data mismatch and needs to be fixed in Firestore.*

TEST 2: Call History (All Calls)

`GET http://62.171.170.48:4500/api/v1/calls`

Header:

```
Authorization: Bearer ck_live_d3188b50_94a44765acfc97d19ced4416
```

Expected Response:

```
{
  "calls": [...],
  "pagination": { "page": 1, "limit": 20, "total": 47, "totalPages": 3 }
}
```

What it returns: All completed + active calls for this account, sorted newest first.

> *If `calls` is empty but `ownedAgents > 0` — calls exist but have no `userId` field. This is the older call data issue. New calls made after the restart will appear correctly.*

TEST 3: Active Calls (Live Right Now)

`GET http://62.171.170.48:4500/api/v1/supervisor/calls`

Header:

```
Authorization: Bearer ck_live_d3188b50_94a44765acfc97d19ced4416
```

Expected Response (during an active call):

```
[
  {
    "id": "abc123-def456",
    "phoneNumber": "+919876543210",
    "agentId": "agent-id-here",
    "agentName": "Priya - Sales",
    "status": "active",
    "sentiment": "neutral",
    "startedAt": "2026-03-17T13:30:00.000Z"
  }
]
```

Expected Response (no active calls):

```
[]
```

> *Empty array `[]` means no one is currently on a live call. This is correct — not an error.*

Troubleshooting Guide

Error	Cause	Fix
`{"error": "Invalid or expired token"}`	Server not restarted	Run `git stash && git pull && pm2 restart all`
`{"error": "Missing or invalid Authorization header."}`	Wrong format	Header must be exactly: `Authorization: Bearer ck_live_...`
`{"error": "Invalid or revoked API Key."}`	API key is wrong or revoked	Generate a new key from Dashboard → Settings
`calls: []` (empty, no error)	No calls for this account yet	Make a test call, then check again
`ownedAgents: 0` in identity check	API key belongs to wrong user	Check which account created the API key
`{"error": "Not found"}` on `/api/v1/...`	Server running old code	Restart with `pm2 restart all`
`500 Internal Server Error`	Firestore connection issue	Check `pm2 logs` on server

What Was Fixed (March 17, 2026)

Fix 1 — Call History List Was Returning 404

File: `enterprise/backend/src/routes/analytics.js`

The `GET /api/analytics/calls` endpoint (used by the dashboard UI) did not exist. Added it with:

- Direct userId match
- AgentId fallback for older calls missing userId
- Pagination support

Fix 2 — Call Completion Not Saving Duration

File: `app/main.py`

When a call ended, the Firestore update was using a fragile query to find the document. Fixed to:

- Directly use call UUID as document ID
- Save `duration` field (seconds)
- Always save `userId` on call completion

Fix 3 — Whisper/Barge Returning 404 for Older Calls

File: `enterprise/backend/src/routes/external.js`

Whisper and Barge endpoints were doing strict `call.userId === apiUserId` check. Fixed to also check if the call's `agentId` belongs to this user — so older calls without userId still work.

Quick Command Reference (Server)

```
# Pull latest code
git stash && git pull origin main

# Restart everything
pm2 restart all

# Check all process status
pm2 list

# View live logs (Node.js backend)
pm2 logs enterprise-backend --lines 50

# View live logs (Python voice engine)
pm2 logs callex-python --lines 50

# Health check
curl http://localhost:4500/api/health
```

Postman Quick Setup

1. Open Postman
2. New Collection → "Callex API"
3. Add a **Collection Variable**: `api_key` = `ck_live_d3188b50_94a44765acfc97d19ced4416`
4. In each request, Header: `Authorization` → `Bearer {{api_key}}`

Request Name	Method	URL
Identity Check	GET	`http://62.171.170.48:4500/api/v1/debug/my-identity`
All Call History	GET	`http://62.171.170.48:4500/api/v1/calls`
Active Calls	GET	`http://62.171.170.48:4500/api/v1/supervisor/calls`
List Agents	GET	`http://62.171.170.48:4500/api/v1/agents`

> ***Remember:*** *Every time code is updated on GitHub, you must* `*git pull*` *+* `*pm2 restart all*` *on the server. Without restart, the old code keeps running.*